

	Component Name	Quantity	Specification	Required/Optional	Approx Price (INR)	Purpose
	ESP32-CAM	1	AI Thinker WiFi + Camera Module	Required	800	Main controller and camera
	ESP32-CAM-MB Board OR FTDI Programmer	1	USB Programmer Module	Required	300	Programming and power interface
	DHT22 Sensor	1	Temperature & Humidity Sensor	Required	200	Environmental monitoring
	MQ-2 Gas Sensor	1	Gas and Smoke Detection	Required	100	Detect LPG, smoke, gases
	MQ-135 Air Quality Sensor	1	Air Quality Sensor Module	Required	120	Detect air pollutants
	Flame Sensor	1	IR Flame Detection Module	Required	50	Fire detection
	PIR Motion Sensor	1	HC-SR501 Motion Sensor	Required	120	Motion detection
	Active Buzzer	1	5V Active Buzzer	Required	30	Audio alert system
	Red LED	1	5mm LED	Required	10	Danger indication
	Green LED	1	5mm LED	Required	10	Normal status indication
	220 Ohm Resistor	2	Current limiting resistor	Required	10	Protect LEDs and GPIO
	10k Ohm Resistor	1	Pull-up resistor	Required	5	DHT22 signal stabilization
	1000uF Capacitor	1	Electrolytic capacitor	Recommended	20	Power stability
	Breadboard	1	Standard Breadboard	Required	150	Circuit assembly
	Jumper Wires	40	Male-Male and Male-Female wires	Required	150	Electrical connections
	5V 2A Power Adapter	1	DC Power Supply	Required	200	Provide stable power
	Micro USB Cable	1	USB Cable	Required	100	Programming connection
	MicroSD Card	1	8GB or 16GB Storage	Optional	250	Store captured images
	Relay Module	1	5V Relay Module	Optional	80	Control external devices
	Transistor (BC547)	1	NPN Transistor	Optional	10	Protect buzzer circuit
	Diode (1N4007)	1	Protection diode	Optional	5	Circuit protection
	TOTAL				2720	